

Task 1: SQL Benchmark Dataset Preparation and Evaluation Design

Objective

The goal of this task is to prepare a clean benchmark dataset that will later be used to evaluate your Text-to-SQL agent system.

You will work with a predefined collection of natural language questions and manually create correct SQL queries for them. These SQL queries will act as the **ground truth** (reference answers) for evaluating whether your Text-to-SQL agent is generating correct outputs.

In addition, you will think critically about:

“How do we evaluate whether a Text-to-SQL agent is actually performing well?”

This task forms the foundation for all future assignments in this module.

Dataset

Use the following Google Sheet as the benchmark question dataset:

[SQL Benchmark Questions Dataset](#)

This dataset contains natural language questions related to the provided PostgreSQL database schema.

You will use these questions throughout the remaining assignments in this course.

Instructions

Part 1: Write Ground Truth SQL Queries

For each natural language question in the dataset:

1. Read and understand the question carefully
2. Identify:
 - Tables involved
 - Required joins
 - Filters and conditions

- Aggregations or grouping operations
 - 3. Write a clean and correct SQL query
 - 4. Execute the query in PostgreSQL
 - 5. Verify that the query produces the expected result
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Important Requirements

- Write readable and properly formatted SQL
 - Use aliases where appropriate
 - Avoid unnecessary complexity
 - Ensure all queries are executable
 - Use only safe read-only queries (`SELECT` statements)
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Deliverables for Part 1

You must submit:

- The original natural language question
 - Your final SQL query
 - Query output/result screenshot or exported result
 - Brief explanation of how the query works
 - Prepare a document for this as a submission file where you will attach each question, sql query from you and then the results you have obtained (screenshot),if your result is about selecting all rows of a column, you can show count and few rows in the screenshot.
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Part 2: Design an Evaluation Strategy for Text-to-SQL Agents

Objective

Now think beyond query generation.

You are building an AI agent that:

- Understands natural language
- Generates SQL
- Executes queries
- Returns answers

The question is:

“How do we know if the agent is actually good?”

Your Task

Research and propose methods to evaluate a Text-to-SQL system.

You are encouraged to think critically and explore possible evaluation approaches.

Things to Consider

You may evaluate the system based on:

- SQL correctness
 - Execution success
 - Accuracy of returned results
 - Correct table and column selection
 - Join correctness
 - Error handling capability
 - Retry/self-correction performance
 - Natural language answer quality
 - Query execution time
 - Robustness to ambiguous questions
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Prepare a document on evaluation benchmark or framework on how we can evaluate such systems and use these evaluation benchmarks in your mini text2sql agent assignment (task 3 and 4)